Cartesian Equation of a Plane

Knowledge/ Understanding

1. For each of the following, find the scalar equation of the plane that passes through the point P_0 and has normal $\vec{n}$.

a. $P_0(2, 1, -3)$, $\vec{n} = (7, 1, -1)$

b. $P_0(5, 1, 9)$, $\vec{n} = (1, 0, 0)$

c. $P_0(0, 6, -2)$, $\vec{n} = (2, 0, 3)$

d. $P_0(0, 0, 0)$, $\vec{n} = (2, -1, 4)$

Knowledge/ Understanding

2. Determine the scalar equation of the plane that passes through $(1, -2, 3)$ and has a normal

a. parallel to the y-axis

b. perpendicular to the xy-plane

c. parallel to the normal of the plane $x - y - 2z + 19 = 0$

Communication

3. a. Find the scalar equation of the plane that passes through the origin and has a normal $\vec{n} = (A, B, C)$.

b. How can you tell by inspection of the scalar equation of a plane whether or not the plane passes through the origin?

5. Find the scalar equation of each of the following planes. State which of the planes, if any, are coincident.

a. $\vec{r} = (-8, -1, 8) + s(-5, 1, 4) + t(3, 2, -4)$

b. $\vec{r} = (-2, -2, 5) + s(3, 1, -1) + t(4, 1, -4)$

c. $\vec{r} = (2, 0, 0) + s(0, 4, 0) + t(0, 0, -3)$

d. $\vec{r} = (-8, 2, 0) + s(4, 0, 3) + t(0, -2, -5)$

e. $\vec{r} = (2, -11, -17) + s(0, 5, 13) + t(0, 3, 10)$

f. $\vec{r} = (13, 0, -12) + s(-1, 8, -4) + t(11, 3, -12)$

6. Find the scalar equation of each of the following planes.

a. $x = 4 + 3s - 2t$
$y = 2 + 4s + 4t$
$z = 1 - 2s - 3t$

b. $x = -2t$
$y = 2 - s - 3t$
$z = 5 + 3s$

7. For each of the following, find the scalar equation of the plane that passes through the given points.

a. $(1, 1, -1)$, $(1, 2, 3)$, $(3, -1, 2)$

b. $(2, -2, 4)$, $(1, 1, -4)$, $(3, 1, -6)$

c. $(1, 1, 1)$, $(-1, 1, 1)$, $(2, 1, 2)$

d. $(1, 3, 0)$, $(0, 5, 2)$, $(3, 4, -2)$

8. What is the scalar equation of the plane that contains the x-axis and the point $(4, -2, 1)$?

Application **9.** Find the scalar equation of the plane that contains the intersecting lines

$$\frac{x-2}{1} = \frac{y}{2} = \frac{z+3}{3} \text{ and } \frac{x-2}{-3} = \frac{y}{4} = \frac{z+3}{2}$$

Application **10.** Determine whether the following pairs of planes are coincident, parallel and distinct, or neither.

a. $x + 3y - z - 2 = 0$ and $2x + 6y - 2z - 8 = 0$

b. $2x + y + z - 3 = 0$ and $6x + 2y + 2z - 9 = 0$

c. $3x - 3y + z - 2 = 0$ and $6x - 6y + 2z - 4 = 0$

d. $2x - 4y + 2z - 6 = 0$ and $3x - 6y + 3z - 9 = 0$

11. Find a vector equation for the plane with scalar equation

a. $2x - y + 3z - 24 = 0$ b. $3x - 5z + 15 = 0$

12. Which of the following lines is parallel to the plane $4x + y - z - 10 = 0$? Do any of the lines lie in the plane?

a. $\vec{r} = (3, 0, 2) + t(1, -2, 2)$

b. $x = -3t, y = -5 + 2t, z = -10t$

c. $\frac{x-1}{4} = \frac{y+6}{-1} = \frac{z}{1}$

13. The angle between two planes is defined as the angle between their normals. Determine the angle θ ($0 \leq \theta \leq 90°$), to the nearest degree, between the given planes.

a. $2x + 3y - z + 9 = 0$ and $x + 2y + 4 = 0$

b. $x - y - z - 1 = 0$ and $2x + 3y - z + 4 = 0$

11. Find the distance between

a. the point $(7, 7, -7)$ and the plane $6y - z + 5 = 0$

b. the point $(3, 2, 1)$ and the plane $3x+2y+z=10$

Answers

1. a. $7x + y - z - 18 = 0$ **b.** $x - 5 = 0$ **c.** $2x + 3z + 6 = 0$
d. $2x - y + 4z = 0$ **2. a.** $y + 2 = 0$ **b.** $z - 3 = 0$
c. $x - y - 2z + 3 = 0$ **3. a.** $Ax + By + Cz = 0$ **b.** $D = 0$
5. a. $12x + 8y + 13z = 0$ **b.** $3x - 8y + z - 15 = 0$ **c.** $x - 2 = 0$
d. $3x + 10y - 4z + 4 = 0$ **e.** $x - 2 = 0$
f. $12x + 8y + 13z = 0$; a and f, c and e are coincident
6. a. $4x - 13y - 20z + 30 = 0$ **b.** $9x - 6y - 2z + 22 = 0$
7. a. $11x + 8y - 2z - 21 = 0$ **b.** $x + 3y + z = 0$ **c.** $y - 1 = 0$
d. $6x - 2y + 5z = 0$ **8.** $y + 2z = 0$
9. $8x + 11y - 10z - 46 = 0$ **10. a.** parallel and distinct
b. neither **c.** coincident **d.** coincident
11. a. $\vec{r} = (0, -24, 0) + s(1, 2, 0) + t(0, 3, 1)$
b. $\vec{r} = (0, 0, 3) + s(5, 0, 3) + t(0, 1, 0)$
12. a. parallel to and on the plane **b.** parallel to the plane, not on
c. not parallel **13. a.** 17° **b.** 90° **15. a.** $6x - 4y - 4z - 3 = 0$

Extra 11.a) $\frac{54\sqrt{37}}{37}$ b) $\frac{2\sqrt{14}}{7}$